

MAT21-3

Multivariable Analysis

Gradients, Hessians, and the Calculus Behind Gradient Descent

From the slope of a curve to the geometry of optimization in many dimensions.

Albert School — 22 hours

Preface

Single-variable calculus (MAT11-1) gave you a complete toolkit for functions of one real variable: you can differentiate $f(x)$, read its derivative as a slope, locate extrema by solving $f'(x) = 0$, decide whether a critical point is a minimum or maximum with f'' , and approximate f near a point with a Taylor polynomial. You also met convexity — the property of “lying above every tangent line” — and saw why it forces a local minimum to be the global one.

Almost nothing in modern modelling is a function of a single variable. A machine-learning cost depends on thousands of parameters; a physical field depends on position in space; a profit depends on several prices at once. This course extends every single-variable idea above to functions $f(x_1, \dots, x_n)$ of several variables. The slope $f'(x)$ becomes a **vector**, the **gradient** ∇f ; the second derivative $f''(x)$ becomes a **matrix**, the **Hessian** H_f ; the first- and second-derivative tests become statements about that vector and that matrix.

This is exactly where your linear algebra pays off. From MAT11-3 you bring vectors in $\mathbb{R}^n$, the dot product, and matrix–vector products. From MAT21-1 you bring eigenvalues, eigenvectors, and the idea that a symmetric matrix is understood through its eigenvalues — the precise tool we need to classify a critical point and to explain why some optimization problems are easy and others are slow. The course closes the loop by assembling all of this into the **gradient descent** algorithm and the **Jacobian**, the two objects that Deep Learning Math and Optimization will assume you already own.

A guiding habit throughout: whenever a multivariable statement looks intimidating, **freeze all variables but one**. What remains is an ordinary single-variable function, and your MAT11-1 intuition applies verbatim. We use this trick to define partial derivatives, and we return to it constantly.

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1 Functions of Several Variables

1.1 From one input to several

A function of several variables takes a point of $\mathbb{R}^n$ and returns a single real number. We write

$$f : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}, \quad (x_1, \dots, x_n) \mapsto f(x_1, \dots, x_n).$$

Almost all of our intuition is built in the case $n = 2$, where we can draw pictures, with occasional remarks about $n = 3$ and general n .

Worked example — three everyday two-variable functions.

$$f(x, y) = x^2 + y^2, \quad g(x, y) = xy, \quad h(x, y) = \sqrt{1 - x^2 - y^2}.$$

The first measures squared distance to the origin; the second is a product that will become our standard **saddle**; the third is the upper half of the unit sphere, defined only where $x^2 + y^2 \leq 1$.

1.2 Domains

The **domain** of f is the set of inputs for which the formula makes sense. We treat domains **operationally** — by inspecting which operations are legal — not through any formal topology.

The two recurring obstructions. A domain in $\mathbb{R}^2$ or $\mathbb{R}^3$ is almost always carved out by one of two rules you already know from MAT11-1:

- a square root requires its argument to be ≥ 0 ;
- a denominator (or the argument of a logarithm) must be nonzero (resp. > 0).

Worked example — reading a domain off a formula. For $h(x, y) = \sqrt{1 - x^2 - y^2}$ we need $1 - x^2 - y^2 \geq 0$, i.e. $x^2 + y^2 \leq 1$: the closed unit disk. For $k(x, y) = \ln(x - y)$ we need $x - y > 0$, i.e. the open half-plane strictly below the line $y = x$.

Common pitfall A domain in $\mathbb{R}^2$ is a region of the **plane**, not an interval. “Sketch the domain” means shade the set of valid points (x, y) , including a clear indication of whether each boundary curve is included (solid) or excluded (dashed).

1.3 Graphs and level sets

There are two complementary ways to picture $f : \mathbb{R}^2 \rightarrow \mathbb{R}$.

The **graph** is the surface $z = f(x, y)$ sitting in $\mathbb{R}^3$. It is faithful but hard to draw and harder to read off the page.

The **level set** (or **level curve**, **contour**) at height c is

$$L_c = \{(x, y) \in D : f(x, y) = c\}.$$

It is the curve in the plane along which f is constant — exactly the contour lines of a topographic map. A whole family of level curves, labelled by their heights, is called a **contour plot** and is usually the most informative single picture of a two-variable function.

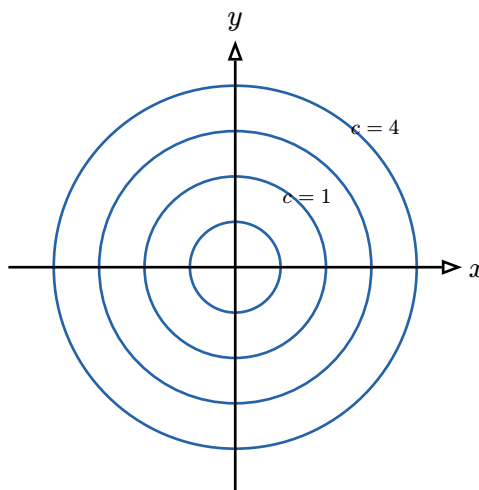


Figure 1: Contour plot of $f(x, y) = x^2 + y^2$. Each level curve $x^2 + y^2 = c$ is a circle of radius $\sqrt{c}$. Circles drawn closer together signal a steeper slope.

Remark For $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ the analogue of a level curve is a **level surface** $f(x, y, z) = c$. For example the level surfaces of $f(x, y, z) = x^2 + y^2 + z^2$ are spheres centred at the origin.

Exercises

1. Sketch the domains of (a) $\sqrt{xy}$, (b) $1/(x^2 - y)$, (c) $\ln(4 - x^2 - y^2)$.
2. Describe and sketch the level curves of $f(x, y) = y - x^2$. What shape are they, and how are they spaced?
3. For $g(x, y) = xy$, draw the level curves $g = 1$, $g = -1$, and $g = 0$. (The last is a degenerate “curve” — explain.)
4. Name the level surfaces of $f(x, y, z) = x^2 + y^2 - z$.

2 Partial Derivatives and Differentiability

2.1 Partial derivatives: freeze all but one variable

To differentiate a function of several variables we use the habit announced in the preface: hold every variable fixed except one, and differentiate the ordinary one-variable function that remains.

Definition 1 (partial derivative) Let $f : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ and (a, b) an interior point of D . The **partial derivative of f with respect to x** at (a, b) is

$$\frac{\partial f}{\partial x}(a, b) = \lim_{h \rightarrow 0} \frac{f(a + h, b) - f(a, b)}{h},$$

when the limit exists; the partial derivative with respect to y is defined symmetrically. We also write f_x and f_y .

In practice you never compute the limit: you differentiate with the usual MAT11-1 rules, treating the other variable as a constant.

Worked example — mechanical computation. For $f(x, y) = x^2y + \sin(x) + y^3$:

$$f_x = 2xy + \cos(x), \quad f_y = x^2 + 3y^2.$$

In f_x the term y^3 is a constant and vanishes; in f_y the term $\sin(x)$ is a constant and vanishes.

2.2 Geometric meaning: slopes of cross-sections

Fixing $y = b$ slices the graph $z = f(x, y)$ with the vertical plane $y = b$. The intersection is an ordinary curve $z = f(x, b)$, and $f_{x(a,b)}$ is its slope at $x = a$. Likewise $f_{y(a,b)}$ is the slope of the cross-section in the y -direction. A partial derivative is the slope you feel when you walk along the surface parallel to one of the axes.

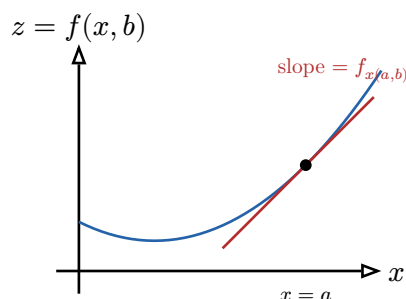


Figure 2: Slicing the graph by the plane $y = b$ leaves the curve $z = f(x, b)$. Its slope at $x = a$ is the partial derivative $f_{x(a,b)}$.

2.3 Existence of partials is not enough

In one variable, differentiability and “having a derivative” are the same thing. In several variables they come apart, and this is the single most important conceptual hazard of the chapter.

Definition 2 (differentiability) f is **differentiable** at (a, b) if there is a vector (A, B) such that, as $(h, k) \rightarrow (0, 0)$,

$$f(a + h, b + k) = f(a, b) + Ah + Bk + o(\|(h, k)\|).$$

The linear part $Ah + Bk$ is the **best linear approximation** to f near (a, b) ; when it exists one necessarily has $A = f_{x(a,b)}$ and $B = f_{y(a,b)}$.

Differentiability says the graph is well approximated by a **tangent plane** in **every** direction at once. Merely having f_x and f_y only controls the two axis directions, which is strictly weaker.

Common pitfall Partials can exist while f is not even continuous. The classic example is

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}.$$

Along the axes f is identically 0, so $f_{x(0,0)} = f_{y(0,0)} = 0$ both exist. But along the line $y = x$ we get $f(x, x) = \frac{x^2}{2x^2} = \frac{1}{2}$ for all $x \neq 0$, so f does not tend to $f(0, 0) = 0$: it is **discontinuous** at the origin. Existence of partials is genuinely weaker than differentiability.

2.4 The practical criterion: $C^1 \Rightarrow$ differentiable

We rarely check the definition directly. The following theorem is the working tool: it converts the hard property (differentiability) into an easy check (continuity of the partials).

Theorem 3 (C^1 criterion) If f_x and f_y exist and are **continuous** on an open set around (a, b) , then f is differentiable at (a, b) . We say f is of class C^1 .

This gives the chain of implications you should memorise — together with the fact that **none of the arrows reverses**:

$$f \in C^1 \implies f \text{ differentiable} \implies f \text{ continuous} \quad \text{and} \quad f \text{ differentiable} \implies f_x, f_y \text{ exist.}$$

One-line summary. Continuous partials $\implies$ differentiable $\implies$ continuous, and differentiable $\implies$ partials exist. The converses all fail; the counterexample above kills the rightmost one. Since every function built from polynomials, exponentials, logs, and trig by the usual operations has continuous partials on its domain, in practice such functions are differentiable everywhere they are defined.

Exercises

5. Compute both partial derivatives of (a) $f(x, y) = x^3y^2 - 4xy$, (b) $g(x, y) = e^{xy}$, (c) $h(x, y) = \ln(x^2 + y^2)$.
6. For $f(x, y, z) = xyz + x^2z$, compute f_x, f_y, f_z .
7. Explain in one sentence why $f(x, y) = x^2 + 3y$ is differentiable at every point without checking the limit definition.
8. For the discontinuous example in the pitfall box, verify directly that $f_{x(0,0)} = 0$ using the limit definition, and that the limit of f along $y = 2x$ is not 0.

3 The Gradient and Directional Derivatives

3.1 Assembling the gradient

The two partial derivatives are naturally packaged into one vector.

Definition 4 (gradient) The **gradient** of a differentiable f at (a, b) is the vector of its partial derivatives,

$$\nabla f(a, b) = \begin{pmatrix} f_{x(a,b)} \\ f_{y(a,b)} \end{pmatrix} \in \mathbb{R}^2,$$

and analogously $\nabla f = (f_{x_1}, \dots, f_{x_n})$ in $\mathbb{R}^n$.

With the gradient, the best linear approximation from the previous chapter becomes a single dot product:

$$f(a + h, b + k) \approx f(a, b) + \nabla f(a, b) \cdot \begin{pmatrix} h \\ k \end{pmatrix}.$$

Worked example — a gradient as a vector field. For $f(x, y) = x^2 + y^2$ we have $\nabla f = (2x, 2y)$. At $(1, 1)$ the gradient is $(2, 2)$; at $(-1, 0)$ it is $(-2, 0)$. Every gradient vector points straight away from the origin — the direction of fastest increase of “squared distance”.

3.2 Directional derivatives

Partial derivatives measure the rate of change along the axes only. To measure the rate along an **arbitrary** direction we use the gradient.

Definition 5 (directional derivative) Let $\mathbf{u}$ be a **unit** vector ($\|\mathbf{u}\| = 1$). The **directional derivative** of f at $\mathbf{p}$ in the direction $\mathbf{u}$ is

$$D_{\mathbf{u}}f(\mathbf{p}) = \lim_{t \rightarrow 0} \frac{f(\mathbf{p} + t\mathbf{u}) - f(\mathbf{p})}{t}.$$

Theorem 6 (gradient formula for the directional derivative) If f is differentiable at $\mathbf{p}$ then for every unit vector $\mathbf{u}$,

$$D_{\mathbf{u}}f(\mathbf{p}) = \nabla f(\mathbf{p}) \cdot \mathbf{u}.$$

This is the payoff of differentiability: a single vector, ∇f , encodes the slope in **every** direction at once, retrieved by a dot product.

Common pitfall The formula $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$ requires $\mathbf{u}$ to be a **unit** vector. If you are handed a direction $\mathbf{v}$ that is not normalised, first replace it by $\mathbf{u} = \mathbf{v}/\|\mathbf{v}\|$. Forgetting to normalise is the most common error here and silently rescales your answer.

3.3 Steepest ascent

Write the dot product with its angle form (MAT11-3):

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = \|\nabla f\| \|\mathbf{u}\| \cos \theta = \|\nabla f\| \cos \theta,$$

where θ is the angle between $\mathbf{u}$ and ∇f . This single line reads off three facts at once.

The gradient is the compass of steepest ascent.

- $D_{\mathbf{u}}f$ is largest when $\theta = 0$: the steepest **increase** is in the direction of ∇f , with rate $\|\nabla f\|$.
- It is most negative when $\theta = \pi$: the steepest **decrease** is in the direction $-\nabla f$. **This is the direction gradient descent will move in.**
- It is zero when $\theta = \pi/2$: moving **perpendicular** to ∇f keeps f instantaneously constant — the seed of the next chapter.

Worked example — where to step. For $f(x, y) = x^2 + y^2$ at $(1, 1)$, $\nabla f = (2, 2)$. To increase f fastest, step along $(2, 2)$ (i.e. the unit vector $(1, 1)/\sqrt{2}$); the rate is $\|(2, 2)\| = 2\sqrt{2}$. To decrease f fastest, step along $(-2, -2)$ — toward the origin, exactly where this bowl bottoms out.

Exercises

9. Compute ∇f for (a) $f = x^2y$, (b) $f = e^x \sin y$, (c) $f = x/(x + y)$.
10. For $f(x, y) = x^2 + 3xy$, find the directional derivative at $(1, 2)$ in the direction of $\mathbf{v} = (3, 4)$. (Remember to normalise.)
11. At $(1, 0)$ for $f = x^2 + y^2$, in which direction does f increase fastest, and what is that fastest rate?
12. Find a direction $\mathbf{u}$ along which $D_{\mathbf{u}}f(1, 2) = 0$ for $f = x^2 + y^2$.

4 Chain Rule, Level Curves, and Tangent Planes

4.1 The chain rule along a curve

Suppose a point moves through the plane along a parametrised curve $\mathbf{r}(t) = (x(t), y(t))$, and we watch the value $g(t) = f(\mathbf{r}(t))$. How fast does f change as we travel?

Theorem 7 (chain rule along a curve) If f is differentiable and $x(t), y(t)$ are differentiable, then

$$\frac{d}{dt}f(x(t), y(t)) = f_x \cdot x'(t) + f_y \cdot y'(t) = \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t).$$

The compact form — **gradient dotted with velocity** — is worth memorising: the rate of change of f along a path is the dot product of its gradient with the path's velocity vector.

Worked example — heating along a path. Let $f(x, y) = x^2 + y^2$ and $\mathbf{r}(t) = (\cos t, \sin t)$, a point circling the unit circle. Then $\nabla f = (2x, 2y) = (2 \cos t, 2 \sin t)$ and $\mathbf{r}'(t) = (-\sin t, \cos t)$, so

$$g'(t) = (2 \cos t)(-\sin t) + (2 \sin t)(\cos t) = 0.$$

The value never changes — of course: the path is a level curve of f .

4.2 The gradient is orthogonal to level curves

That example was no accident; it is the general theorem, and the proof is a one-line application of the chain rule.

Theorem 8 (gradient orthogonality) Let $\mathbf{r}(t)$ be a curve lying inside a level set $f = c$. Then at every point of the curve,

$$\nabla f \cdot \mathbf{r}'(t) = 0.$$

That is, ∇f is orthogonal to the level curve's tangent direction.

Proof. On the curve, $f(\mathbf{r}(t)) = c$ is constant. Differentiating both sides with the chain rule gives $\nabla f \cdot \mathbf{r}'(t) = \frac{d}{dt}(c) = 0$. Since $\mathbf{r}'(t)$ is tangent to the level curve, ∇f is perpendicular to it. $\square$

This dovetails with the previous chapter: moving **along** a level curve keeps f constant, so the directional derivative there is 0, so $\mathbf{u} \perp \nabla f$. The gradient always points **across** the contours, in the steepest direction.

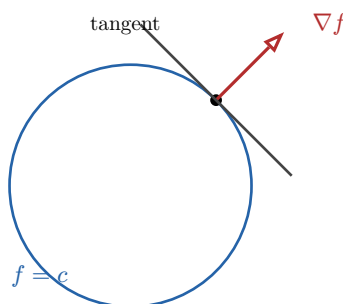


Figure 3: The gradient ∇f at a point is perpendicular to the level curve through that point, pointing toward higher values.

4.3 Tangent lines and tangent planes

Orthogonality gives equations for tangent objects almost for free, because a line (or plane) is pinned down by a point on it and a normal vector to it.

Proposition 9 (tangent line to a level curve) The tangent line to the level curve $f(x, y) = c$ at (a, b) has equation

$$f_{x(a,b)}(x - a) + f_{y(a,b)}(y - b) = 0,$$

since $\nabla f(a, b)$ is normal to the curve.

Proposition 10 (tangent plane to a graph) For the graph $z = f(x, y)$, the tangent plane at $(a, b, f(a, b))$ is

$$z = f(a, b) + f_{x(a,b)}(x - a) + f_{y(a,b)}(y - b).$$

This is precisely the best linear approximation of Chapter 2 written as a plane.

Worked example — a tangent plane. For $f(x, y) = x^2 + y^2$ at $(1, 2)$: $f(1, 2) = 5$, $f_x = 2x = 2$, $f_y = 2y = 4$. The tangent plane is

$$z = 5 + 2(x - 1) + 4(y - 2) = 2x + 4y - 5.$$

Common pitfall Keep the two pictures apart. The tangent **line** lives in the plane and is the linearisation of a **level curve** $f = c$ (a 2D object, normal ∇f in $\mathbb{R}^2$). The tangent **plane** lives in space and is the linearisation of the **graph** $z = f(x, y)$ (a 3D object). They use the same partials but answer different questions.

Exercises

13. For $f = x^2 - y^2$ and $\mathbf{r}(t) = (t, t^2)$, compute $\frac{d}{dt}f(\mathbf{r}(t))$ two ways: by the chain rule, and by substituting first then differentiating.
14. Find the tangent line to the level curve $x^2 + 4y^2 = 5$ at $(1, 1)$.
15. Find the tangent plane to $z = xy$ at $(2, 3)$.
16. Verify, for $f = x^2 + y^2$ and the circle $\mathbf{r}(t) = (3 \cos t, 3 \sin t)$, that $\nabla f \cdot \mathbf{r}'(t) = 0$ at every t .

5 The Hessian and Second-Order Taylor Expansion

5.1 Second-order partial derivatives

Each partial derivative f_x and f_y is itself a function of (x, y) and can be differentiated again. This produces four **second-order partials**:

$$f_{xx} = \frac{\partial}{\partial x}(f_x), \quad f_{xy} = \frac{\partial}{\partial y}(f_x), \quad f_{yx} = \frac{\partial}{\partial x}(f_y), \quad f_{yy} = \frac{\partial}{\partial y}(f_y).$$

The **mixed** partials f_{xy} and f_{yx} differentiate with respect to both variables, in the two possible orders.

Worked example — computing the four. For $f(x, y) = x^2y + y^3$: first $f_x = 2xy$ and $f_y = x^2 + 3y^2$. Then

$$f_{xx} = 2y, \quad f_{xy} = 2x, \quad f_{yx} = 2x, \quad f_{yy} = 6y.$$

Notice $f_{xy} = f_{yx}$ — no coincidence, as the next theorem explains.

5.2 Schwarz's theorem: order does not matter

Theorem 11 (Schwarz / Clairaut) If the mixed partials f_{xy} and f_{yx} exist and are continuous on an open set around (a, b) (in particular if f is C^2), then

$$f_{xy}(a, b) = f_{yx}(a, b).$$

The order of differentiation is irrelevant.

Common pitfall The hypothesis is **continuity** of the mixed partials, not their mere existence. There exist (pathological) functions with $f_{xy}(0, 0) \neq f_{yx}(0, 0)$; in those the mixed partials are discontinuous at the origin. For every function you will meet in this course the hypothesis holds, so you may freely assume symmetry — but cite the hypothesis when asked.

5.3 The Hessian matrix

Schwarz's theorem says the four second partials really carry only three distinct numbers, arranged in a **symmetric** matrix.

Definition 12 (Hessian) The **Hessian** of f at $\mathbf{p}$ is the matrix of second-order partials

$$H_{f(\mathbf{p})} = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}.$$

By Schwarz's theorem it is symmetric: $f_{xy} = f_{yx}$.

Symmetry is precisely what makes the Hessian tractable: a real symmetric matrix has real eigenvalues and an orthonormal basis of eigenvectors (MAT21-1). We will classify critical points by the **signs** of those eigenvalues.

Worked example — a Hessian. For $f(x, y) = x^2y + y^3$ at the point $(1, 1)$, using the partials above,

$$H_{f(1,1)} = \begin{pmatrix} 2y & 2x \\ 2x & 6y \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 6 \end{pmatrix}.$$

5.4 Second-order Taylor expansion

In one variable, $f(a + h) \approx f(a) + f'(a)h + \frac{1}{2}f''(a)h^2$. The multivariable version replaces f' by the gradient and f'' by the Hessian.

Theorem 13 (second-order Taylor expansion) For a C^2 function f and a small displacement $\mathbf{h} = (h, k)$,

$$f(\mathbf{p} + \mathbf{h}) = f(\mathbf{p}) + \nabla f(\mathbf{p}) \cdot \mathbf{h} + \frac{1}{2} \mathbf{h}^\top H_{f(\mathbf{p})} \mathbf{h} + o(\|\mathbf{h}\|^2).$$

The middle term is the linear approximation; the quadratic term $\frac{1}{2} \mathbf{h}^\top H_f \mathbf{h}$ is the leading correction, controlled entirely by the Hessian.

At a **critical point**, where $\nabla f = 0$, the linear term vanishes and the quadratic term is the local shape of f — which is exactly why the Hessian decides whether a critical point is a minimum, a maximum, or a saddle. That is the subject of the next chapter.

Exercises

17. Compute all four second partials of (a) $f = x^3 + xy^2$, (b) $f = e^x \cos y$, and verify Schwarz's theorem in each case.
18. Write the Hessian of $f = x^2 + xy + y^2$ at an arbitrary point. Is it constant?
19. Write the second-order Taylor expansion of $f = \sin(x) \cos(y)$ at the origin.
20. Give the Hessian of $f(x, y, z) = x^2 + y^2 + z^2$ (a 3×3 matrix).

6 Classifying Critical Points

6.1 Critical points

Definition 14 (critical point) A point $\mathbf{p}$ is a **critical point** of f if $\nabla f(\mathbf{p}) = \mathbf{0}$, i.e. $f_{x(\mathbf{p})} = f_{y(\mathbf{p})} = 0$ simultaneously.

As in one variable (Fermat, MAT11-1), every interior local extremum is a critical point — at a smooth peak or valley the tangent plane is horizontal. But a critical point need not be an extremum: it may be a **saddle**, a point that is a minimum along one direction and a maximum along another.

Worked example — the three model shapes. All three have a single critical point at the origin, where $\nabla f = 0$:

- $f = x^2 + y^2$: a **minimum** (a bowl);
- $f = -x^2 - y^2$: a **maximum** (a dome);
- $f = x^2 - y^2$: a **saddle** — rising along x , falling along y .

6.2 The Monge criterion

To classify a critical point without plotting, examine the Hessian. The classical two-variable test, due to Monge, uses the entries directly. Write

$$r = f_{xx}, \quad s = f_{xy}, \quad t = f_{yy}, \quad D = rt - s^2 = \det H_f.$$

Theorem 15 (Monge criterion) Let $\mathbf{p}$ be a critical point of a C^2 function and $D = rt - s^2$ the determinant of $H_{f(\mathbf{p})}$.

- If $D > 0$ and $r > 0$: **local minimum**.
- If $D > 0$ and $r < 0$: **local maximum**.
- If $D < 0$: **saddle point**.
- If $D = 0$: **inconclusive** — the test fails; investigate by other means.

Worked example — classifying a saddle. Let $f(x, y) = x^2 - y^2$. Then $\nabla f = (2x, -2y) = 0$ only at the origin. There $r = 2$, $s = 0$, $t = -2$, so $D = (2)(-2) - 0 = -4 < 0$: a **saddle**, as expected.

Worked example — a worked classification. Let $f(x, y) = x^3 - 3x + y^2$. From $\nabla f = (3x^2 - 3, 2y) = 0$ we get $y = 0$ and $x = \pm 1$: two critical points $(1, 0)$ and $(-1, 0)$. Here $f_{xx} = 6x$, $f_{yy} = 2$, $f_{xy} = 0$.

- At $(1, 0)$: $r = 6 > 0$, $t = 2$, $D = 12 > 0$ — local **minimum**.
- At $(-1, 0)$: $r = -6$, $t = 2$, $D = -12 < 0$ — **saddle**.

6.3 The eigenvalue viewpoint

The Monge criterion is a shortcut for a deeper statement about the eigenvalues of the Hessian — the viewpoint that survives into higher dimensions and into the analysis of gradient descent. Recall from the Taylor expansion that near a critical point $f(\mathbf{p} + \mathbf{h}) \approx f(\mathbf{p}) + \frac{1}{2}\mathbf{h}^\top H \mathbf{h}$. The local shape is therefore governed by the quadratic form $\mathbf{h}^\top H \mathbf{h}$, whose sign is read off the eigenvalues of the symmetric matrix H (MAT21-1).

Theorem 16 (eigenvalue classification) Let λ_1, λ_2 be the eigenvalues of $H_{f(\mathbf{p})}$ at a critical point.

- Both $\lambda_i > 0$ (positive definite): **local minimum** — f curves upward in every direction.
- Both $\lambda_i < 0$ (negative definite): **local maximum**.
- Eigenvalues of **opposite sign** (indefinite): **saddle point**.
- Some $\lambda_i = 0$: **inconclusive**.

The link to Monge is exact, because for a 2×2 matrix the determinant and trace are the product and sum of the eigenvalues (MAT21-1):

$$D = \det H = \lambda_1 \lambda_2, \quad \text{tr } H = r + t = \lambda_1 + \lambda_2.$$

So $D > 0$ means the eigenvalues share a sign (definite — extremum), and that shared sign is the sign of r ; $D < 0$ means opposite signs (saddle). The two criteria are one criterion in two languages.

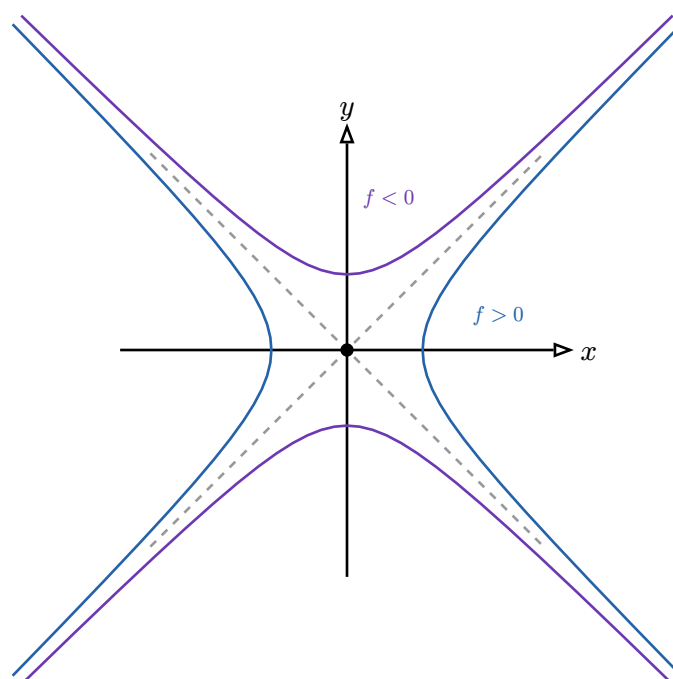


Figure 4: Contours of the saddle $f = x^2 - y^2$. Level curves are hyperbolae; f rises along the x -axis and falls along the y -axis. The eigenvalues of the Hessian (+2 and -2) have opposite signs.

Common pitfall $\nabla f = 0$ is **necessary** but not **sufficient** for an extremum: a saddle has zero gradient too. Always follow up with the Hessian test. And when $D = 0$ the test says **nothing** — do not conclude “saddle”; the point could be a minimum (e.g. $f = x^4 + y^4$ at the origin), a maximum, or a saddle.

Exercises

21. Find and classify all critical points of $f(x, y) = x^2 + xy + y^2 - 3x$.
22. Find and classify the critical points of $f(x, y) = x^3 + y^3 - 3xy$.
23. Show $f(x, y) = x^4 + y^4$ has $D = 0$ at the origin, yet the origin is a (global) minimum. What does this illustrate?
24. Compute the eigenvalues of the Hessian of $f = x^2 + 4xy + y^2$ at the origin and classify the critical point both ways.

7 Gradient Descent

7.1 The algorithm

We now have everything needed to **minimise** a function numerically. The steepest ascent chapter showed $-\nabla f$ is the direction of fastest decrease; gradient descent simply takes repeated small steps in that direction.

Definition 17 (gradient descent) Choose a starting point $\mathbf{x}_0$ and a **step size** (or **learning rate**) $\alpha > 0$. Iterate

$$\mathbf{x}_{n+1} = \mathbf{x}_n - \alpha \nabla f(\mathbf{x}_n).$$

Each step moves downhill, perpendicular to the local contour. When the gradient is large (steep terrain) the steps are long; as we approach a critical point $\nabla f \rightarrow 0$, the steps shrink and the iteration stalls — ideally at a minimum.

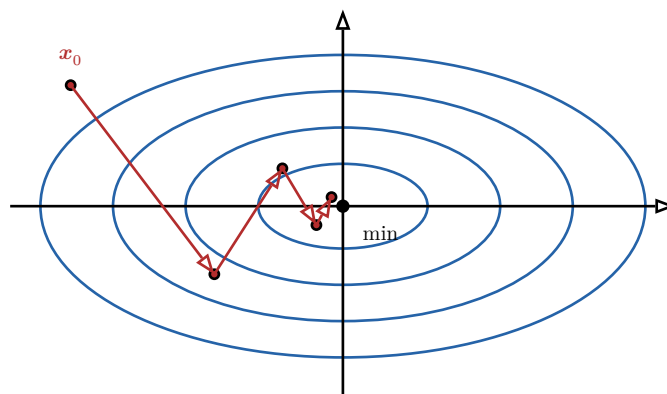


Figure 5: Gradient descent on elliptical contours. Each step is perpendicular to the contour it leaves. On a stretched (ill-conditioned) bowl the path zig-zags toward the minimum instead of heading straight in.

7.2 The role of the step size

The step size α trades speed against stability, and choosing it is the central practical art.

Reading the learning rate.

- **α too small:** every step is safe but tiny; convergence is correct but painfully slow.
- **α too large:** steps overshoot the minimum and the iterates can **oscillate** or even **diverge**, with $f(x_n)$ increasing.
- **α well chosen:** steady, rapid decrease of $f(x_n)$.

Worked example — a one-dimensional sanity check. Minimise $f(x) = x^2$, where $\nabla f = 2x$, so $x_{n+1} = x_n - 2\alpha x_n = (1 - 2\alpha)x_n$. This is a geometric sequence (MAT11-1) with ratio $1 - 2\alpha$. It converges to 0 precisely when $|1 - 2\alpha| < 1$, i.e. $0 < \alpha < 1$; at $\alpha = 1/2$ it reaches the minimum in a single step; for $\alpha > 1$ it diverges. The whole story of the learning rate, in one line.

7.3 Convergence toward critical points

Gradient descent moves downhill, so under mild conditions $f(x_n)$ decreases and the iterates approach a point where $\nabla f = 0$. Two honest caveats:

Common pitfall Gradient descent finds **critical points**, and in practice **local** minima — not necessarily the global minimum, and not, by itself, a saddle-avoiding path. The direction $-\nabla f$ only knows the local slope. Whether the critical point it reaches is the global minimum is a separate question, answered by convexity in the next chapter.

7.4 Conditioning: why eigenvalues set the speed

Near a minimum, the second-order Taylor expansion says f looks like the quadratic bowl $\frac{1}{2}\mathbf{h}^\top H\mathbf{h}$, whose contours are ellipses with axes along the Hessian's eigenvectors and **widths set by its eigenvalues**. This is what couples the Hessian to the **speed** of gradient descent.

Definition 18 (condition number) For a symmetric positive-definite Hessian with eigenvalues $0 < \lambda_{\min} \leq \dots \leq \lambda_{\max}$, the **condition number** is

$$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}}.$$

Why ill-conditioning is slow. The eigenvalue λ_i is the **curvature** along its eigenvector: large λ means a steep, narrow direction, small λ a shallow, flat one. A single step size α must serve both. To stay stable, α must be small enough for the **steepest** direction ($\lambda_{\max}$); but that same α makes progress along the **flattest** direction ($\lambda_{\min}$) agonisingly slow. The mismatch is measured by κ :

- $\kappa \approx 1$ (circular bowl): descent heads almost straight to the minimum, fast.
- $\kappa \gg 1$ (long thin valley): descent **zig-zags** across the valley, as in the figure above, and convergence slows roughly in proportion to κ .

Worked example — a stretched bowl. For $f(x, y) = x^2 + 100y^2$, the Hessian is $\begin{pmatrix} 2 & 0 \\ 0 & 200 \end{pmatrix}$ with eigenvalues 2 and 200, so $\kappa = 100$. Stability in the y -direction caps α near $1/200$, but then the x -direction (curvature 2) creeps in with effective ratio $1 - 2\alpha \approx 0.99$ per step — hundreds of steps to make real progress. Rescaling the variables to make the bowl round ($\kappa = 1$) would fix it, and is the elementary idea behind **preconditioning**.

Exercises

25. Run two steps of gradient descent on $f(x, y) = x^2 + y^2$ from $\mathbf{x}_0 = (2, 2)$ with $\alpha = 0.1$. What is $\mathbf{x}_2$?
26. For $f(x) = x^2$ and $\alpha = 0.6$, compute $x_1, \dots, x_4$ from $x_0 = 1$. Does it converge? Relate your answer to the ratio $1 - 2\alpha$.
27. For $f(x, y) = x^2 + 9y^2$, give the Hessian, its eigenvalues, and the condition number. Predict qualitatively how the descent path behaves.
28. Explain in one or two sentences why a step size that works on $x^2 + y^2$ may diverge on $x^2 + 100y^2$.

8 Local vs Global Extrema and Convexity

8.1 The gap between local and global

Everything so far is **local**: the Hessian test and gradient descent both reason from behaviour near a point. But minimising a model means finding the **global** minimum — the lowest value anywhere on the domain.

Definition 19 (local vs global minimum) $\mathbf{p}$ is a **local minimum** of f if $f(\mathbf{p}) \leq f(\mathbf{x})$ for all $\mathbf{x}$ in some neighbourhood of $\mathbf{p}$. It is a **global minimum** if the inequality holds for all $\mathbf{x}$ in the domain. Maxima are defined symmetrically.

In general a function has many local minima of differing heights, and no local test can certify that the one you found is global. We need extra structure — and the right structure is convexity, which you met in one variable (MAT11-1).

8.2 Convex functions

Definition 20 (convex function) f is **convex** on a convex domain if for all points $\mathbf{x}, \mathbf{y}$ and all $t \in [0, 1]$,

$$f(t\mathbf{x} + (1-t)\mathbf{y}) \leq tf(\mathbf{x}) + (1-t)f(\mathbf{y}).$$

Geometrically: the graph lies **below** every chord (equivalently, **above** every tangent plane) — the multivariable echo of “lying above its tangent lines”.

For a C^2 function there is a clean second-order test, exactly parallel to “ $f'' \geq 0$ ” in one variable, with the Hessian playing the role of f'' .

Theorem 21 (second-order convexity test) A C^2 function f is convex on an open convex set if and only if its Hessian $H_{f(\mathbf{x})}$ is positive semidefinite (all eigenvalues ≥ 0) at every point.

8.3 Why convexity is the dividing line

Theorem 22 (convexity makes local global) If f is convex, then every local minimum of f is a **global** minimum. Moreover any critical point ($\nabla f = 0$) is a global minimum. If f is strictly convex, the global minimum is unique.

This is the theorem that makes optimization tractable. For a convex cost, gradient descent’s caveat evaporates: the local minimum it finds **is** the global one, and a single critical point is the answer. For a non-convex cost (a generic deep network), no such guarantee holds, which is why initialisation, step-size schedules, and randomness matter so much in practice.

Worked example — a convex bowl. $f(x, y) = x^2 + y^2$ has Hessian $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$, positive definite everywhere, so f is (strictly) convex. Its only critical point, the origin, is therefore the unique global minimum — no further checking required.

8.4 A simple machine-learning cost

The tools of this course were built for exactly the following calculation, which is the skeleton of **linear least-squares** / a one-neuron model.

Worked example — least squares as a convex minimisation. Given data points (t_i, y_i) , fit a line $y = wt + b$ by minimising the mean squared error over the two parameters (w, b) :

$$J(w, b) = \frac{1}{m} \sum_{i=1}^m (wt_i + b - y_i)^2.$$

This is a function of two variables, w and b , and we apply the full pipeline:

1. **Gradient.** Differentiating term by term,

$$J_w = \frac{2}{m} \sum_i (wt_i + b - y_i)t_i, \quad J_b = \frac{2}{m} \sum_i (wt_i + b - y_i).$$

2. **Critical point.** Setting $\nabla J = 0$ gives two linear equations in (w, b) — the **normal equations** — whose solution is the best-fit line.

3. **Classification.** The Hessian is

$$H_J = \frac{2}{m} \begin{pmatrix} \sum t_i^2 & \sum t_i \\ \sum t_i & m \end{pmatrix},$$

a constant, positive-semidefinite matrix (it is positive definite as soon as the t_i are not all equal). So J is **convex**, the critical point is the **global** minimum, and there are no spurious local minima.

4. **Gradient descent.** Because J is convex with constant Hessian, gradient descent

$$(w, b) \leftarrow (w, b) - \alpha \nabla J$$

converges to that global minimum; its speed is governed by the condition number of H_J , which is why practitioners **standardise** the inputs t_i (making H_J well-conditioned) before training.

This single example exercises every idea in the course — domain, gradient, critical point, Hessian, convexity, global optimality, and gradient-descent convergence — on the object machine learning actually cares about.

Exercises

29. Show $f(x, y) = e^x + e^y$ is convex by computing its Hessian. Where is its minimum?
 30. Give a function of one variable with a local minimum that is not global, and explain why your example is not convex.
 31. For the least-squares cost $J(w, b)$ with data $(1, 1), (2, 2), (3, 2)$, write the normal equations $\nabla J = 0$ and solve for (w, b) .
 32. Show that $f(x, y) = x^2 - y^2$ is **not** convex by exhibiting a direction in which the Hessian's quadratic form is negative.

9 Vector-Valued Functions: the Jacobian

(general knowledge)

9.1 When the output is a vector

So far the output of f has been a single number. Many maps return a **vector**: a coordinate change, a layer of a neural network, the right-hand side of a system of differential equations. Such a map is

$$\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad \mathbf{F}(\mathbf{x}) = \begin{pmatrix} F_1(\mathbf{x}) \\ \vdots \\ F_m(\mathbf{x}) \end{pmatrix},$$

with m scalar component functions $F_1, \dots, F_m$, each of n variables.

9.2 The Jacobian matrix

The right first-order object is no longer a single gradient but a **stack** of gradients — one row per output component.

Definition 23 (Jacobian) The **Jacobian** of $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ at $\mathbf{x}$ is the $m \times n$ matrix of all first-order partials,

$$J_{\mathbf{F}}(\mathbf{x}) = \begin{pmatrix} \frac{\partial F_1}{\partial x_1} & \cdots & \frac{\partial F_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial F_m}{\partial x_1} & \cdots & \frac{\partial F_m}{\partial x_n} \end{pmatrix}.$$

Its i -th row is the gradient ∇F_i (transposed) of the i -th component.

Remark Two familiar objects are special cases. If $m = 1$ (scalar output), the Jacobian is a single row — the gradient written sideways. If $\mathbf{F}$ is **linear**, $\mathbf{F}(\mathbf{x}) = A\mathbf{x}$, then $J_{\mathbf{F}} = A$ everywhere: the Jacobian is the natural generalisation of “the matrix of a linear map” (MAT11-3) to nonlinear maps, holding only locally.

Worked example — a Jacobian. For $\mathbf{F}(x, y) = (x^2y, x + \sin y)$ from $\mathbb{R}^2$ to $\mathbb{R}^2$,

$$J_{\mathbf{F}} = \begin{pmatrix} \frac{\partial}{\partial x}(x^2y) & \frac{\partial}{\partial y}(x^2y) \\ \frac{\partial}{\partial x}(x + \sin y) & \frac{\partial}{\partial y}(x + \sin y) \end{pmatrix} = \begin{pmatrix} 2xy & x^2 \\ 1 & \cos y \end{pmatrix}.$$

9.3 The chain rule as a product of Jacobians

The reason the Jacobian is **the** right object is that it turns the multivariable chain rule into ordinary matrix multiplication — the operation you drilled in MAT11-3.

Theorem 24 (multivariable chain rule) If $\mathbf{G} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $\mathbf{F} : \mathbb{R}^m \rightarrow \mathbb{R}^p$ are differentiable, then the composition $\mathbf{F} \circ \mathbf{G}$ is differentiable and its Jacobian is the **matrix product** of the Jacobians, evaluated at the matching points:

$$J_{\mathbf{F} \circ \mathbf{G}}(\mathbf{x}) = J_{\mathbf{F}}(\mathbf{G}(\mathbf{x})) \cdot J_{\mathbf{G}}(\mathbf{x}).$$

The single-variable chain rule $(f \circ g)' = f'(g(x)) \cdot g'(x)$ is the 1×1 case; the chain-rule-along-a-curve of Chapter 4 is the case where $\mathbf{G} : \mathbb{R} \rightarrow \mathbb{R}^n$ is a path, so $J_{\mathbf{G}} = \mathbf{r}'(t)$ and the matrix product becomes the dot product $\nabla f \cdot \mathbf{r}'$.

Why this is the course’s exit door. This statement — **the derivative of a composition is the product of the derivative matrices** — is the engine of **backpropagation** in deep learning: a network is a long composition of layers, and the gradient of its loss is computed by multiplying the layers’ Jacobians from output back to input. Optimization courses likewise assume it as background. You are not expected to prove it here; you are expected to **state** it, recognise its shape, and know that the Jacobian is the multivariable derivative that makes it work.

Common pitfall Matrix multiplication does not commute (MAT11-3), and neither does the Jacobian chain rule: the **order** $J_{\mathbf{F}} \cdot J_{\mathbf{G}}$ matters, and the dimensions must match ($p \times m$ times $m \times n$ gives $p \times n$). Writing the product in the wrong order is the standard slip — the outer map’s Jacobian comes **first**, on the left.

Exercises

- 33.** Compute the Jacobian of $\mathbf{F}(x, y) = (x + y, xy, x - y)$ from $\mathbb{R}^2$ to $\mathbb{R}^3$. What are its dimensions?
- 34.** For $\mathbf{F}(x, y) = (e^x \cos y, e^x \sin y)$, compute $J_{\mathbf{F}}$ and its determinant.
- 35.** Let $\mathbf{G}(t) = (t^2, t^3)$ and $f(x, y) = x + y$. Compute the Jacobian of $f \circ \mathbf{G}$ two ways: by the chain-rule product, and by substituting first.
- 36.** State, with correct dimensions, the chain rule for $\mathbf{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ composed after $\mathbf{G} : \mathbb{R}^2 \rightarrow \mathbb{R}^3$.